

Jahnvi R

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ACADEMIC HISTORY

PES University, Bachelors in Technology - Computer Science Engineering

2023 – 2027 (expected)

EXPERIENCE

MTRDC -DRDO (Microwave Tube Research and Development Centre - Defense Research and Development Centre)

06/2025 – 07/2025

INTERN

Centralized Logging Infrastructure – rsyslog (4 Clients, 1 Server)

- Set up a secure centralized logging system using rsyslog across 5 Linux machines.
- Configured log reception via UDP/TCP and organized client logs with custom templates.
- Ensured reliable log forwarding, connectivity, and delivery.

System & Network Information Gathering Tool – Bash Script

- Developed an interactive, menu-driven Bash script to audit system metrics and perform network reconnaissance.
- Implemented pure Bash IP manipulation for CIDR-based scanning without external tools.

Machine Learning - Teaching Assistant

06//2026 – present

PROJECTS

SkillSwap - Peer-to-Peer Learning Platform 

Java, Spring Boot, Spring Security, JWT, JPA/Hibernate, PostgreSQL, H2, JavaFX, Maven

- Developed a full-stack peer-learning platform with a Spring Boot backend and JavaFX client, supporting doubt posting, answers, feedback, upvotes, and skill-based discovery.
- Designed RESTful APIs and a relational persistence layer using JPA/Hibernate with PostgreSQL, following controller-service-repository architecture.
- Implemented JWT-based authentication and Spring Security, providing secure access to application resources and user-specific functionality.

ATLAS-KV – Fault-Tolerant Distributed Key-Value Store 

Java, Spring Boot, Raft Consensus, REST APIs, Docker, Write-Ahead Logging (WAL)

- Engineered a distributed key-value store using the Raft consensus algorithm, supporting leader election, replicated logs, and majority-quorum writes across multiple nodes.
- Implemented automatic leader failover and fault-tolerant replication to maintain consistency and availability during node failures.
- Built durable persistence using Write-Ahead Logging (WAL) and containerized the multi-node cluster with Docker for reproducible deployment.

BuildingDNA – Autonomous Building Intelligence System 

Python, EnergyPlus, Ollama, LLMs, MCP, Streamlit, Pandas

- Built an autonomous building-control system integrating EnergyPlus simulations with local LLM reasoning and a safety-validated closed-loop architecture.
- Reduced electricity consumption by 8.73% and estimated comfort violations by 61.47% against a matched simulation baseline.

Autonomous Competitive Intelligence Agents 

Python, FastAPI, Qdrant, LLMs, Vector Search, Pydantic

- Developed a multi-agent pipeline for parallel retrieval, extraction, verification, and semantic storage using FastAPI and Qdrant.
- Implemented confidence scoring, source verification, and conflict detection to improve reliability of generated intelligence.

TECHNICAL SKILLS

- **Languages:** Java, C++, Python, C , SQL, JavaScript, Bash
- **Backend & APIs:** Spring Boot, FastAPI, REST APIs, Spring Security, JWT, JPA/Hibernate
- **Databases & Distributed Systems:** PostgreSQL, MySQL, H2, Qdrant, Raft Consensus, Replication, Leader Election, Write-Ahead Logging (WAL)
- **Cloud & DevOps:** Docker, Git, GitHub, Linux, GCP
- **Core CS:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP
- **AI/ML:** PyTorch, Hugging Face, LangGraph, LLMs, RAG, Vector Search, Pandas, NumPy, Scikit-learn

ACHIEVEMENTS AND LEADERSHIP

The Entrepreneurship Cell

- Member of Board of Directors

Nexus (Tech Club)

- Design Head

<https://github.com/jahnvi1504>

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